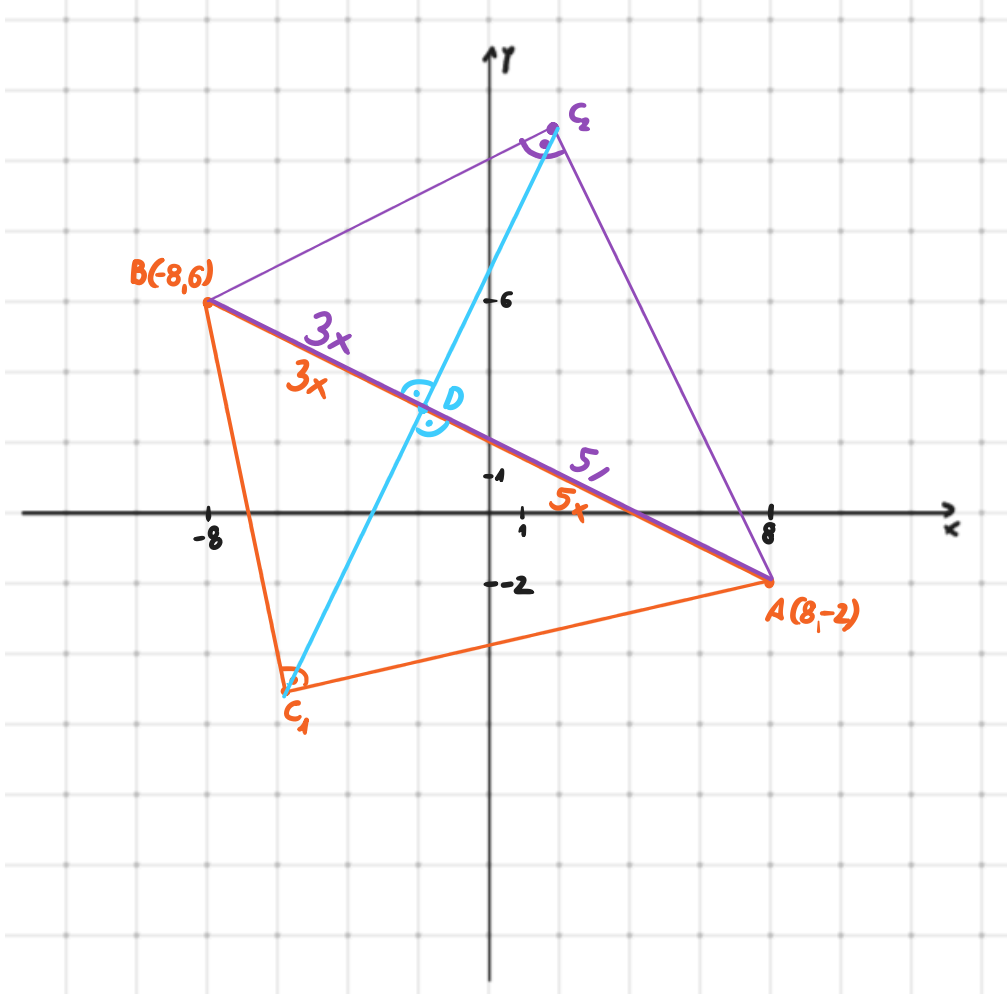


1) Tworzymy rysunek poglądowy:


$$\overrightarrow{AB} = [-8-8, 6+2] = [-16, 8]$$

$$\vec{AD} = \frac{5}{8} \vec{AB} = [-10, 5]$$

$$[-10, 5] = [x_D - 2, y_D + 2]$$

$$x_D = -2, y_D = 3 \quad D(-2, 3)$$

4) Wyznaczamy punkt C z tw. Pitagorasa:

$$\begin{aligned} \text{pr. CD: } y &= ax + b \\ \text{pr. AB} \perp \text{pr. CD} &\Leftrightarrow a_{AB} \cdot a_{CD} = -1 \\ a_{CD} &= 2 \end{aligned}$$

$$D(-2, 3) \in \text{pr. CD} \Leftrightarrow 3 = 2 \cdot (-2) + b$$
$$b = 7$$

pr. CD: $y = 2x + 7$

$$\text{pr. AB: } y = -\frac{1}{2}x + 2$$

$$C(x, 2x+7)$$
$$|AC|^2 + |BC|^2 = |AB|^2$$
$$|AC| = \sqrt{(x-8)^2 + (2x+7+2)^2}, |BC| = \sqrt{(x+8)^2 + (2x+7-6)^2}, |AB| = \sqrt{(8-3)^2 + (-2-6)^2} = \sqrt{320} = 8\sqrt{5}$$

$$\begin{aligned}(x-8)^2 + (2x+9)^2 + (x+8)^2 + (2x+1)^2 &= 320 \\ (x^2 - 16x + 64) + (4x^2 + 36x + 81) + (x^2 + 16x + 64) + (4x^2 + 4x + 1) &= 320 \\ 10x^2 + 40x + 210 &= 320 \\ 10x^2 + 40x - 110 &= 0 \quad | :10 \\ x^2 + 4x - 11 &= 0 \\ \Delta = 16 - 4(-11) = 60, \quad \sqrt{\Delta} = 2\sqrt{15}\end{aligned}$$

$$x_1 = \frac{-4 - 2\sqrt{5}}{2} = -2 - \sqrt{5} \Rightarrow y_1 = 2 \cdot (-2 - \sqrt{5}) + 7 = 3 - 2\sqrt{5}$$

$$x_2 = -2 + \sqrt{5} \Rightarrow y_2 = 2 \cdot (-2 + \sqrt{5}) + 7 = 3 + 2\sqrt{5}$$

Odp. $C = (-2 - \sqrt{15}, 3 - 2\sqrt{15})$ lub $C = (-2 + \sqrt{15}, 3 + 2\sqrt{15})$.